

SENIOR AI SYSTEMS ARCHITECT & PLATFORM ENGINEER

Senior Systems Architect and Platform Engineer with 20+ years of hands-on experience designing, implementing, and deploying enterprise-grade architectures, local LLM API servers, high-performance backend pipelines, and security-centric crypto platforms. Expert in localized inference deployment (quantized GGUF models, CUDA layer offloading, context orchestration), custom retrieval-augmented generation (RAG) vector engines, multi-agent automated systems, and high-performance WebSockets systems.

PROFESSIONAL EXPERIENCE

Adarma & Global Clients (Remote)

Senior AI Systems Architect & Platform Consultant

Oct 2024 – Present

- Architected and deployed private local LLM inference engines using llama.cpp and custom GPU offloading wrappers, handling context windows up to 128K tokens.
- Designed self-hosted RAG architectures utilizing ChromaDB and sentence-transformer embeddings to securely index large-scale codebase semantics.
- Built custom multi-agent workflow runners mapping filesystems, auditing code schemas, and executing autonomous telemetry feedback loops.
- Configured enterprise-grade cloud inference configurations utilizing AWS Bedrock with robust service fallback handlers and token consumption policies.

IronNet & Global Enterprises (Remote)

Senior Systems Engineer & Backend Developer

Dec 2020 – Sep 2024

- Engineered high-throughput asynchronous telemetry pipelines and API endpoints using FastAPI, Celery, and Redis, decreasing mean processing latency by 35%.
- Developed multi-tenant dashboard systems deployed on Cloudflare Workers CDN, serving global regions with sub-450ms data syncing.
- Supervised migration, replication, and indexing strategies for high-volume PostgreSQL database clusters with zero service interruptions.
- Created real-time monitoring environments using Prometheus & Grafana with alert policies triggered by network anomalies.

Rubica & SMB Portfolios (Remote)

Full Stack & Infrastructure Engineer

Feb 2018 – Dec 2020

- Designed and maintained scalable web platform structures for 20+ corporate and SMB clients, deploying secure backend structures and responsive UI portals.
- Hardened REST APIs with Zero-Trust access control, JWT authentication, and AES-256 state encryption.
- Automated infrastructure provisioning and containerization using Docker, Docker Compose, and shell workflows.

CORE COMPETENCIES

AI & LLM ORCHESTRATION

- Quantized Local Models (GGUF)
- llama.cpp & CUDA Offloading
- RAG & Vector Search Engines
- Multi-Agent Autonomous Loops
- Context Window Architecture

SYSTEMS & PLATFORM

- FastAPI / Python / Node.js
- WebSockets & Live Telemetry
- Cloudflare Workers Edge API
- PostgreSQL, Redis, Supabase
- Asynchronous Task Workers

SECURITY & CRYPTOGRAPHY

- Zero-Knowledge Proofs (ZKP)
- WebAuthn & FIDO2 Protocols
- Zero-Trust API Hardening
- Threat Intelligence & OSINT
- BTCPay Server Integrations

CORE TECH STACK

- Python FastAPI TypeScript React
Node.js Cloudflare Workers Supabase
PostgreSQL Redis Docker Celery
llama.cpp WebSockets CUDA ZKP
Git Linux

PROFESSIONAL EXPERIENCE (CONT.)

Upwork — Top Rated Plus (100% Job Success Rate)
Senior AI, Cybersecurity & Full-Stack Consultant Apr 2017 – Present

- Successfully completed over 50 projects for clients in North America, Europe, and Middle East.
- Delivered customized APIs, complex database schemas, real-time WebSockets solutions, and vector search integrations.
- Maintained 100% Client Satisfaction Score with continuous repeat business engagement.

FLAGSHIP INITIATIVES

VaultGuard — Institutional P2P Crypto Desk

TypeScript · Cloudflare Workers · Supabase · React · WebAuthn

Production-deployed institutional OTC settlement engine running across Cloudflare's 300+ city edge network. Integrates Zero-Knowledge Proof progressive identity vaults, passwordless WebAuthn/FIDO2 biometrics, and threat intelligence ingestion (VirusTotal, AbuseIPDB, Shodan).

C2 Server — Autonomous AI Agent Command Loop

Python · FastAPI · WebSockets · Redis · Docker · Ollama

Next-generation command & control dashboard integrating local LLM orchestration. Directs autonomous scraping, environment mapping, and CVE lookup operations with real-time feedback loops.

Mythos Learner — Repository Indexing Agent

Python · Ollama · ChromaDB · sentence-transformers

Local autonomous background runner designed to map directories, scan config schemas, build unified semantic indexes, and adapt platform tools.

CERTIFICATIONS

- HuggingFace — AI Agents Course
- DeepLearning.AI — Fundamentals of LLMs
- Microsoft / FCC — Foundational C#
- IBM — System & Network Security
- IBM — Incident Response & Forensics
- IBM — Cloud Security Architecture
- IBM — Vulnerability Management
- IBM — GRC & Data Privacy

LANGUAGES

English: Proficient (IELTS 8.0+) Arabic: Fluent (IELTS 7.0+) Persian: Native

Hotel Animator Coordinator
Summers 2008 – 2011 · Multilingual guest operations

Heavy Vehicle Operator (Cat. C+E)
Commercial Logistics · 2005 – 2010

Archaeological Field Practitioner
Mediterranean field research · Tunisia